VERIFICATION CERTIFICATE FOR THE LEAN 4 DEVELOPMENT
P2Q-JUMPS-LEAN

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Abstract. This document certifies that the Lean 4 formalization in the repository `p2q-jumps-lean` builds with its pinned toolchain, contains no `sorry` and no user-declared axioms, and that every theorem depends only on Lean's standard axioms.

1. Verification certificate

The formal development was machine-checked as follows.

- Repository:
  `https://github.com/BenFrohman/p2q-jumps-lean`
- Verified commit:
  `821e35e305306d0dd33a35926d52d21fa5b99a58`
- Lean toolchain: `leanprover/lean4:v4.26.0`
- Mathlib: tag `v4.26.0`, commit (from `lake-manifest.json`)
  `2df2f0150c275ad53cb3c90f7c98ec15a56a1a67`
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  `https://github.com/BenFrohman/p2q-jumps-lean/actions/runs/41568060001`
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  `https://github.com/BenFrohman/p2q-jumps-lean/archive/821e35e305306d0dd33a35926d52d21fa5b99a58.tar.gz`
  `30a3fcdf575403bd0011674814504057d8e2a76880b1689eb20c59675f8ae363`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 4 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`P2qJumps.g_ge_six`	`[propext, Classical.choice, Quot.sound]`
`P2qJumps.g_sub_six_eq`	`[propext, Classical.choice, Quot.sound]`
`P2qJumps.quad_sos`	`[propext, Classical.choice, Quot.sound]`
`P2qJumps.theorem28_from_paper`	`[propext, Classical.choice, Quot.sound]`

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